# PARTH MISHRA

## AI & Systems Engineer • Python Backend & LLM Architecture

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## SUMMARY

B.Tech CSE (AI/ML) candidate at VIT Bhopal (CGPA: 8.67) and **Oracle Certified in Agentic AI**. Engineered an in-memory vector database from first principles in Python (HNSW graph indexing, 38x faster than brute force) and production RAG pipelines grounded on local Ollama models. Experienced in building high-throughput FastAPI/Python backends, event-driven multi-agent systems, and streaming LLM integrations. 250+ LeetCode problems solved. Targeting AI Engineer & Backend SDE roles.

## TECHNICAL SKILLS

- **Languages:** Python (Proficient), C++, TypeScript, JavaScript (ES6+), SQL (PostgreSQL, MySQL), HTML5/CSS3
- **Generative AI & LLMs:** Agentic AI (LangGraph, MCP), RAG Architectures, Vector Search (HNSW, KD-Trees), Claude API, Groq, Ollama, Prompt Engineering, Semantic Embeddings, Structured Outputs
- **Backend & Systems:** FastAPI, Flask, Pydantic v2, RESTful APIs, Node.js, Express.js, Concurrency, Rate Limiting (SlowAPI), JWT Authentication, Row-Level Locking, Idempotency
- **DevOps & Infrastructure:** Docker, Docker Compose, Git/GitHub, Linux/Unix CLI, Postman, Vercel, Render, CI/CD Workflows
- **Core CS & Math:** Data Structures & Algorithms (250+ LeetCode), Linear Algebra, Probability & Statistics, PCA, Distributed Systems Principles

## FEATURED PROJECTS

**VectorDB — In-Memory Vector Search Engine & RAG Pipeline | *Python, Flask, NumPy, Ollama, HNSW* • [GitHub]**
- Architected an in-memory vector database from scratch in NumPy; implemented Hierarchical Navigable Small World (HNSW) graph indexing and KD-Trees with Cosine and Euclidean metrics.
- Slashed nearest-neighbor retrieval latency to **sub-15ms across 50,000+ 768-dimensional embeddings**, delivering a **38x throughput speedup** over baseline brute-force search.
- Built an end-to-end local RAG pipeline with document chunking, semantic search, Ollama (Llama 3.2) grounding, and interactive 2D PCA cluster visualization.

**Forge — Autonomous Multi-Agent Job Hunting & Telemetry System | *Python, Playwright, SQLite, Anthropic Claude, FastAPI* • [GitHub]**
- Engineered an autonomous 4-agent workflow (Scout, Evaluator, CV Adapter, Dispatcher) utilizing token-efficient zero-shot rule filtering, dropping 95%+ irrelevant jobs prior to model inference.
- Implemented dual-resume ATS semantic routing, automated form-filling via headless Playwright, and real-time IMAP telemetry logging 110+ applications into SQLite.

**AI Portfolio & Interactive Recruiter Intelligence Agent | *React, Vite, FastAPI, Groq API, SlowAPI* • [Live: parthh.runs-on.dev] • [GitHub]**
- Shipped high-concurrency streaming recruiter assistant on FastAPI and Groq API (openai/gpt-oss-20b), grounded on candidate project telemetry to prevent hallucinations.
- Implemented anti-abuse rate limits (20 req/min chat, 5 req/min JD matching) with SlowAPI; automated resume suitability scoring and skill gap analysis.

**AI Meal Tracker & Vision Intelligence Engine | *React, Vite, Claude API, Vision LLMs, Prompt Engineering* • [GitHub]**
- Developed an autonomous nutrition intelligence app converting free-text/image food descriptions into strictly structured JSON macronutrient breakdowns via Claude Vision API.
- Engineered domain-specific prompt priors encoding Indian regional culinary portion sizes, boosting macro accuracy over generic USDA reference databases.

## EXPERIENCE

**Software Engineering Intern (Front-End & Systems) — Craftedge Academy | *Remote / Bhopal, India • Oct 2025 – Apr 2026***
- Architected modular, responsive UI modules in React and JavaScript across client-facing academy platforms serving 1,000+ active learners with 100% QA pass rate.
- **Reduced initial page load latency by 28%** via component code-splitting, lazy-loading, and asset minification; eliminated mobile layout shifts.
- **Cut redundant network queries by 35%** without regressions by implementing centralized client-side state caching and debounced REST API hooks.

## EDUCATION

**Vellore Institute of Technology (VIT Bhopal)** — B.Tech in Computer Science & Engineering (AI/ML) • **CGPA: 8.67 / 10.0** • *2023 – 2027*
*Core Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Natural Language Processing, DBMS, Operating Systems, Computer Networks*

## CERTIFICATIONS & ACHIEVEMENTS

- **Oracle Certified Foundations Associate — Agentic AI** (Oracle University, Sept 2026) • [Verify] | Autonomous Agent Architectures & LLM Orchestration
- **CS50's Introduction to Artificial Intelligence with Python** (Harvard University, Oct 2026) • [Verify] | Minimax, Search Algorithms, Neural Networks & NLP
- **Applied Machine Learning in Python** (University of Michigan / Coursera, Jan 2026) • [Verify] | Scikit-Learn Pipelines, Feature Engineering & Model Evaluation
- **Competitive Programming:** Solved **250+ algorithmic problems** across LeetCode, GFG, HackerRank (Dynamic Programming, Graph Theory, Trees, Two Pointers).
- **Academic Honors:** Ranked Top 3 in District Mathematics Olympiad (1,200+ candidates); Top 5 in State Mathematics Olympiad (7,000+ candidates).